

1. **Driver/vehicle owner login:** As a driver/vehicle owner, I want a secure login that will allow me to save my preferred payment, my license plate, car details, reservation history.
2. **Real time parking:** As a driver/vehicle owner, I should be able to see what parking is available
3. **Pricing:** As a driver/vehicle owner, I would like to be able to compare different prices from different parking garages
4. **Parking details:** As a driver/vehicle owner, I would like to see if parking garages amenities such as ADA parking availability or if the parking garage is secure. I would also like to see smart tags like “shortest walk” and “best value”
5. **Notification:** As a driver/vehicle owner and an app user, I would like to receive a notification ahead of time to know when my reservation is ending. With the app I would like to receive an in app notification. With no app I would like to receive either a text message or an email.
6. **Search:** As a driver/vehicle owner, I should be able to search the event that I am attending and find parking garages near it.
7. **Filtering:** As a driver/vehicle owner, I should be able to filter what amenities I want for a parking garage such as valet parking and EV charging.
8. **Navigation:** As a driver/vehicle owner, I would like to receive the address of the parking garage
9. **Mobile Parking pass:** As a driver/vehicle owner, I would like to receive email or message for parking pass and have the ability to save it on google/apple wallet. It can also be available on the app.
10. **Map:** As a driver/vehicle owner, I should have access to a map when looking through parking garages.
11. **Add time:** As a mobile user, if I am running late I would like to have the ability to add time to my reservation
12. **Dashboard:** As a facility manager, I want a real-time dashboard showing reserved vs. drive-up capacity so that I can activate dynamic surge pricing when the lot reaches 80% capacity.
13. **Check in on the attendant side:** As a parking facilities operator, I want to scan a QR code on a customer's phone so that I can instantly verify their reservation without manually cross-referencing a printed list.
14. **License Plate checking:** As an enforcement officer, I want to have a system that validates license plates that belong that have made reservations.
15. **Cancellations:** As a driver/vehicle owner, I would like to be able to cancel my reservation and know about the cancelling policies as well.
16. **Refunds:** As a billing specialist, I want an automated process for issuing partial refunds when events or drivers cancel so that I don't have to manually process thousands of individual transactions.
17. **Login resets:** As a system admin, I would like to be able to restore user login and issues

18. **Handling traffic flow:** As a traffic planner, I want an aggregated heat map of parking utilization during major events so that I can optimize road closures and direct traffic flow more efficiently.
19. **Daily reporting:** As a finance manager, I want automated daily settlement reports detailing total revenue minus the platform's processing fees so that I can easily reconcile with our bank deposits.
20. **Payment:** As a third party app (Apple Pay/Google Pay), I want tokenized payment integration so that the parking app never stores raw credit card numbers on its servers.
21. **Map tokens:** As an admin, I must make sure that API tokens from Google Maps and Apple maps are properly implemented to protect user information.
22. **Role-based access control:** As a security administrator, I want role-based access control (RBAC) configured for the operator dashboard so that a lot attendant cannot accidentally delete the facility's pricing rules.
23. **Manual Entry:** As a lot attendant, I want a manual text-entry field for reservation IDs so that I can still admit vehicles if my device's QR scanner hardware or camera interface fails.
24. **Training Mode:** As a lot attendant, I want a "training mode" toggle in the operator app so that I can practice scanning and validating mock QR codes without affecting live capacity data or revenue.
25. **Latency Alerts:** As a sysadmin, I want API latency alerts integrated directly with PagerDuty so that critical checkout timeouts are escalated immediately if I miss the initial email or dashboard notification.
26. **Read-replicas:** As a database administrator, I want read-replicas configured for real-time spot availability queries so that the primary database does not crash when dynamic surge pricing triggers a massive influx of traffic.
27. **Dashboard for refunds:** As a billing specialist, I want a fallback dashboard utility to batch-process failed refunds so that I do not have to manually execute hundreds of individual transactions if the automated webhook drops.
28. **Margin Calculator:** As a finance manager, I want a margin-calculator tool built into the dynamic pricing engine so that I can ensure third-party platform processing fees never cause us to operate at a loss during heavily discounted hours.
29. **Heat Maps:** As a city official, I want contextual tooltips and a clear legend embedded directly into the utilization heat map so that I do not misinterpret the data when planning municipal road closures.